Deep Parameterized Neural Networks for Hyperspectral Image Denoising

Fengchao Xiong, *Member, IEEE*, Jun Zhou, *Senior Member, IEEE*, Jiantao Zhou, *Senior Member, IEEE*, Jianfeng Lu, *Member, IEEE*, and Yuntao Qian, *Senior Member, IEEE*

***Abstract*— Sparse representation (SR)-based hyperspectral image (HSI) denoising methods normally average the local denoising results of multiple overlapped cubes to recover the whole HSI. Though interpretable, they rely on cumbersome hyperparameter settings and ignore the relationship between overlapped cubes, leading to poor denoising performance. This article combines SR and convolutional neural networks and introduces a deep parameterized sparse neural network (DPNet-S) to address the above issues. DPNet-S parameterizes the SR-based HSI denoising model with two modules: 1) sparse optimizer to extract sparse feature maps from noisy HSIs via recurrent usage of convolution, deconvolution, and soft shrinkage operations; and 2) image reconstructor to recover the denoised HSI from its sparse feature maps via deconvolution operations. We further replace the soft shrinkage operator with U-Net architecture to account for general HSI priors and more effectively capture the complex structures of HSIs, resulting in DPNet-U. Both networks directly learn the parameters from data and perform denoising on the whole HSI, which overcomes the limitations of SR-based methods. Moreover, our networks are generated from the denoising model and optimization procedures, thus leveraging the knowledge embedded and relying less on the number of training samples. Extensive experiments on both synthetic and real-world HSIs show that our DPNet-S and DPNet-U achieve remarkable results when compared with state-of-the-art methods. The codes will be publicly available at https://github.com/bearshng/dpnets for reproducible research.**

***Index Terms*— Convolutional neural networks, hyperspectral image (HSI) denoising, learning to optimize (L2O), sparse representation (SR).**

Manuscript received 1 July 2023; revised 11 August 2023; accepted 12 September 2023. Date of publication 22 September 2023; date of current version 3 October 2023. This work was supported in part by the Natural Science Foundation of Jiangsu Province under Grant BK20200466; in part by the National Natural Science Foundation of China under Grant 62002169, Grant 62371237, Grant 61971476, and Grant 62071421; and in part by the Macao Young Scholars Program under Grant AM2021020. *(Corresponding author: Jiantao Zhou.)*

Fengchao Xiong is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China, and also with the State Key Laboratory of Internet of Things for Smart City, Department of Computer and Information Science, University of Macau, Macau, China (e-mail: fcxiong@njust.edu.cn).

Jun Zhou is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia (e-mail: jun.zhou@griffith.edu.au).

Jiantao Zhou is with the State Key Laboratory of Internet of Things for Smart City, Department of Computer and Information Science, University of Macau, Macau, China (e-mail: jtzhou@umac.mo).

Jianfeng Lu is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: lujf@njust.edu.cn).

Yuntao Qian is with the College of Computer Science, Zhejiang University, Hangzhou 310027, China (e-mail: ytqian@zju.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3318001

I. Introduction

HYPERSPECTRAL image (HSI) has enabled many practical applications, for example, remote sensing [1], face recognition [2], and object tracking [3], [4] due to its material identification ability enabled by numerous continuous bands. Because of the imaging environment, sensor limitation, and imaging mechanism, the captured HSIs are usually degraded by noises, limiting their utility in downstream applications. HSI denoising suffers from the difficulties of 3-D spatial–spectral structure preservation, high-dimensional data, and scarcity of training samples. Moreover, as shown in Fig. 1, the noise variances among bands are always different in real-world HSIs, i.e., non-independent and identically distributed (non-i.i.d.) noise. It makes the HSI denoising problem even more troublesome to handle a large range of noises. Therefore, designing an effective HSI denoising model is challenging but indispensable.

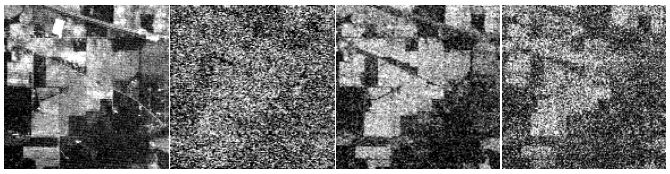
As a typical model-based method, sparse representation (SR) can effectively capture the low-dimensional structure of high-dimensional HSI [5] and has been successfully applied to HSI denoising [6], [7]. SR-based methods assume that the clean HSI is a linear combination of a few components in a prespecified dictionary to filter out the noises. However, because of the curse of dimensionality, most SR-based methods follow the pipeline of imposing HSI priors on local data cubes, extracting all possible cubes, restoring each cube independently, and averaging overlapping pixels between each cube to recover the clean HSI globally. Such an "overlap-averaging" denoising mechanism ignores dependencies between cubes and results in a "local–global" gap and visual artifacts [8].

Deep learning (DL)-based methods, such as convolutional neural networks (CNNs) and transformers [9], [10], perform denoising by learning a nonlinear mapping from noisy–clean HSI pairs. Given sufficient training data, DL-based methods can even perform HSI denoising when the noise model is unknown [11]. However, empirically designed network architectures ignore the problem structure specified for HSI denoising, i.e., mathematical structures such as SR models. As a result, tremendous parameters and large-scale data are required to achieve the desired performance (see Fig. 2).

In order to address the above issues, we replace the key components of the optimization of the L_1 SR model with learnable CNNs, resulting in the deep parameterized sparse neural network (DPNet-S). As shown in Fig. 3, DPNet-S

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(a) (b) (c) (d)

Fig. 1. Some typical bands in the real-world Indian Pines dataset. (a) Band 3. (b) Band 108. (c) Band 164. (d) Band 220.

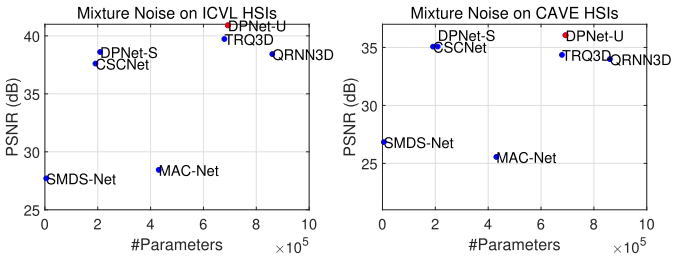


Fig. 2. Our DPNet-S and DPNet-U outperform several state-of-the-art DL-based methods on both ICVL and CAVE datasets with relatively fewer parameters.

includes two modules, i.e., a sparse optimizer and an image reconstructor to mimic the gradient descent for sparse coding and image reconstruction, respectively. On the one hand, compared with the purely data-driven DL-based method, the L_1 SR model helps more effectively utilize the problem structure of HSI denoising and effectively depict the low-dimensional structure of HSIs with fewer parameters. As a result, our DPNet-S is less dependent on the number of training samples. On the other hand, CNN blocks can directly process the whole image, naturally overcoming the drawback of SR-based denoising in cube extraction. In addition, we further replace the sparsity inducer in DPNet-S with the U-Net and propose the DPNet-U method to circumvent the limitation of handcrafted sparse prior by learning the general prior of HSIs from the data. Experimental results on close-range and remote sensing HSIs show that our DPNet-S and DPNet-U outperform both model- and DL-based methods quantitatively and qualitatively.

To summarize, the contributions of this article are threefold.

1) Based on the SR model, we introduce a fully parameterized sparse neural network to capture the low-dimensional structure of HSIs for denoising.
2) We extend DPNet-S to DPNet-U using U-Net to learn the general prior of HSI from the data, which enables to capture the complex structure of HSI more effectively for enhanced denosing.
3) Extensive experiments and thorough analysis confirm the advantages of our methods and show that our methods achieve the state-of-the-art performance.

The following of this article is organized as follows. Section II introduces the related works. The details of the proposed methods are given in Section III. Section IV presents the experimental results. This article concludes in Section V.

II. Related Work

In this section, we introduce related works, including model- and DL-based methods for HSI denoising and learning to optimize (L2O).

A. Model-Based Methods

As HSI denoising is severely underdetermined, model-based methods resort to manually designing different priors depicting the physical structure of the underlying clean HSI to regularize the solution. The strong spectral correlation among bands and the nonlocal spatial similarity drive the success of low-rank models in HSI denoising. In order to obtain the spectral low rankness, some methods employed low-rank matrix factorization [12], [13], [14], [15] or adopted the nuclear norm [16], [17] to directly minimize the rank of a matrix. For example, Zhuang and Bioucas-Dias [13] leveraged the truncated singular value decomposition to depict the low rankness in the spectral domain and used the well-known BM3D denoiser to remove the noises in the spatial domain. Similarly, Peng et al. [18] introduced an iterative manner to refine the subspace and achieved higher denoising performance.

Matrix-based methods unavoidably destroy the spatial–spectral structure of HSIs. Tensor-based methods directly model HSIs under the umbrella of tensor notation and achieve more promising denoising [19], [20], [21]. Besides low rankness, the sparsity prior is also widely applied in HSI denoising. By encoding an HSI with a sparse subset of atoms taken from a redundant dictionary, the noise components can be greatly reduced [6], [22], [23]. Enforcing all the bands to share the same sparse pattern, Ye et al. [6] achieved the spatial–spectral multitask sparse denoising. Moreover, nonlocal self-similarity [24], [25] and a combination of multiple priors [13], [26] have been introduced, yielding more promising denoising.

Model-based methods are interpretable and do not heavily rely on large-scale training samples. However, they usually require computationally expensive optimization iterations and exhaustive hyperparameter tuning to produce high-quality HSI estimation.

B. DL-Based Methods

Instead of implicit modeling, DL-based methods directly learn the nonlinear mapping from the noisy–clean pairs [27], [28]. Early attempts used multichannel 2-D convolutional filters to capture the spatial dependency for HSI denoising [29]. After that, 2-D and 3-D convolutional filters were combined to simultaneously exploit the spatial and spatial–spectral correlations in HSIs [30]. Recently, Wei et al. [31] took advantage of recurrent convolutional networks and 3-D convolutional filters to more effectively capture the global spectral and local spatial–spectral dependencies. Motivated by the flexibility of FFDNet [32] in handling a wide range of noise levels, Maffei et al. [33] introduced a single CNN model for HSI denoising, in which the noise-level map was taken as an input to better balance the denoising performance and preservation of details. Besides CNN-based methods, Transformers have also been adopted in HSI denoising and achieved impressive performance due to their advantages in long-range context exploitation [34]. For example, Li et al. [9] introduced a transformer architecture equipped with the spectral self-attention and window-based spatial self-attention to exploit long-range

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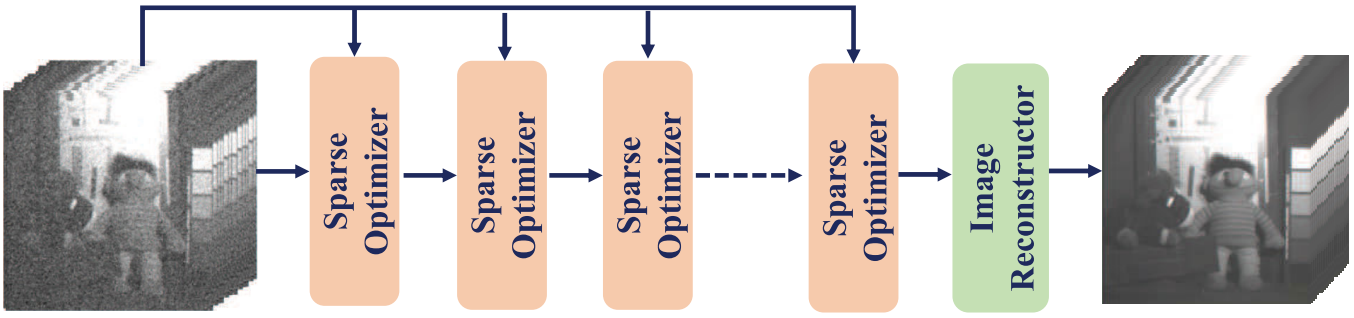


Fig. 3. Given a noisy HSI, we employ a sparse optimizer to produce the sparse feature maps and an image reconstructor to recover the underlying clean HSI from extracted feature maps.

spectral and spatial dependencies respectively. Pang et al. [35] used the window-based multihead self-attention to capture the global spatial features and the locally enhanced feedforward network to exploit the local spatial features.

Most DL-based methods require complicated architectures with a large number of trainable parameters. As a result, they are highly dependent on large-scale training data, which is very costly to acquire in practice. One way is to generate HSIs from RGB images to augment the training dataset [36]. Alternatively, self-supervised methods directly learn a denoiser from the observed noisy image [37], [38], [39]. Zhuang et al. [39] transformed HSI denoising into subspace representation coefficient denoising and generated training pairs from noisy coefficients with neighbor2neighbor strategy [40]. Zhang et al. [41] introduced a self-supervised HSI denoising method, which fine-tunes the DnCNN in the spectral subspace to remove the noises. Combining the physical model can integrate the problem structure of HSI denoising into the network, alleviating the requirements on the number of training samples. Xiong et al. [42] converted the optimizaton of low-rank model aided nonlocal U-Net for HSI denoising. In [43] and [44], low-rank and sparse models were unfolded into a neural network with certain interpretability and achieved high-quality performance.

Following a different line, we parameterize the key steps of the optimization of the SR model with learnable convolutional architectures. Benefiting from the SR model, our network is more capable of capturing the structure of HSIs and requires less data for denoising.

C. Learning to Optimize

Unlike manually designed optimization, "L2O" uses machine learning methods such as DL to learn the target update rules from data [45]. The L2O approach involves designing an optimizer architecture with learnable parameters, which aims to improve the optimization process by parameterizing some key components. During the training process, the current optimizer is iteratively applied to a set of samples. Their performance is then observed, and this information is used to update the parameters of the optimizer. Over time, the optimizer adapts to the training samples, similar to the process of training machine learning models to make predictions or decisions [34]. The learned optimizer not only bypasses the drawbacks of classical optimizers such as laborious iterations and hyperparameter tuning but also can reach better optima with faster speed [34]. In addition, prior knowledge can be embedded to increase the interpretability of DL. The L2O framework also offers an appealing alternative by incorporating data-dependent priors, enabling better adaptation to the actual characteristics of natural signals. Since the proposition of learned iterative shrinkage thresholding algorithm (LISTA) [46], L2O has been introduced to many tasks, such as image deblurring [47] and unmixing [48], [49]. We follow the idea of "L2O" and embed the domain knowledge of HSI denoising to address the limitation of both model- and DL-based methods.

III. Our HSI Denoising Method

This section first briefly revisits the SR model-based HSI denoising and then introduces the proposed deep parameterized networks and their implementations.

A. SR Model for HSI Denoising

Given a clean HSI $\mathcal{X} \in R^{I\times J\times K}$ degraded by zero-mean additive noise $\mathcal{N} \in R^{I\times J\times K}$, the observed noisy HSI $\mathcal{Y} \in R^{I\times J\times K}$ can be modeled by

$$\mathcal{Y} = \mathcal{X} + \mathcal{N}. \tag{1}$$

Recovering $\mathcal{X}$ from $\mathcal{Y}$ is typically an ill-posed inverse problem. Sparsity can effectively regularize this problem and depicts the low-dimensional structure of $\mathcal{X}$ under the assumption that each local cube of HSI is a linear combination of a few atoms in a prespecified dictionary. Accordingly, the problem can be formulated as

$$\min_{\mathbf{s}} \frac{1}{2}\left\|\mathbf{y} - \mathbf{A}\mathbf{s}\right\|_F^2 + \lambda\|\mathbf{s}\|_1 \tag{2}$$

where $\mathbf{y}$ is a reshaped vector version of local cube extracted from $\mathcal{Y}$ and contains $p \times p \times p$ elements, $\mathbf{A} \in \mathbb{R}^{p^3\times m}$ is the dictionary, $\|\cdot\|_1$ is the L_1 norm of $\mathbf{s}$ to promote sparsity, and λ balances the data fidelity term and sparsity promotion term.

Iterative shrinkage thresholding algorithm (ISTA) is a prevailing tool to solve (2) with the following update rule:

$$\mathbf{s}_{k+1} = f_{\text{soft}_{\lambda\alpha}}(\mathbf{s}_k + \alpha(\mathbf{A}^{\mathrm{T}}(\mathbf{y} - \mathbf{A}\mathbf{s}_k))) \tag{3}$$

where α denotes the step size and f_{soft_b} is a soft shrinkage operator defined as $\text{sign}(s)\max(|s| - b, 0)$. The thresholding

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parameter b is associated with hyperparameter λ. Provided with an initialization $\mathbf{s}_0$, running (3) for multiple iterations until convergence yields the solution of $\mathbf{s}$, i.e., $\widehat{\mathbf{s}}$. After that, the clean cube $\mathbf{x}$ is obtained by

$$\mathbf{x} = \mathbf{A}\widehat{\mathbf{s}}. \tag{4}$$

Imposing the local sparsity prior on the whole HSI $\mathcal{X}$, the denoising problem of $\mathcal{Y}$ becomes

$$\min_{\{\mathbf{s}_n\}} \sum_n \left(\lambda \|\mathbf{s}_n\|_1 + \frac{1}{2} \left\| \mathbf{A}\mathbf{s}_n - \mathbf{R}_n \mathcal{Y} \right\|_F^2 \right) \tag{5}$$

where $\mathbf{R}_n$ extracts the nth overlapped cubes from $\mathcal{Y}$. Equation (5) can be solved by first independently computing the SR of each cube with (3) and then estimating the clean HSI $\widehat{\mathcal{X}}$ by averaging pixels between each cube, i.e.,

$$\widehat{\mathcal{X}} = \sum_n \left(\mathbf{R}_n^{\mathrm{T}} \mathbf{R}_n \right)^{-1} \left(\sum_n \mathbf{R}_n^{\mathrm{T}} \mathbf{A}\widehat{\mathbf{s}}_n \right). \tag{6}$$

Unfortunately, the denoising procedure in (3)–(6) usually requires exhaustive iterations and troublesome hyperparameter tuning to obtain high-quality HSI denoising.

B. Proposed Methods

As presented in Fig. 3, the basic idea of DPNet-S is to parameterize some key operations in (3) and (4) with learnable architectures. The former can be modeled as a sparse optimizer using CNNs that consist of a fixed number of layers, while the latter can be modeled as an image reconstructor. Accordingly, we can obtain a mapping from the given noisy HSI to the recovered clean HSI.

1) Sparse Optimizer: In (3), $\mathbf{A}^{\mathrm{T}}$ transforms $\mathbf{y} - \mathbf{A}\mathbf{s}_k$ into the same space with $\mathbf{s}_k$. Since $\mathbf{s}_k$ is the SR of $\mathbf{y}$, $\mathbf{A}^{\mathrm{T}}$ can be regarded as a feature extractor. The role of $\mathbf{A}$ in (3) is to map the representation yielded by the feature extractor to the same space with $\mathbf{y}$ and can be considered as a feature decoder. $f_{\mathrm{soft}_{\lambda\alpha}}(\cdot)$ is the soft shrinkage operator promoting sparsity and can be regarded as a sparsity inducer. The sparse optimizer takes advantage of the powerful representation capability of CNNs as well as its universal approximation property to circumvent the limitations mentioned above of the SR model. Concretely, the optimizer employs convolution and deconvolution operations to parameterize $\mathbf{A}^{\mathrm{T}}$ and $\mathbf{A}$, yielding a universal optimizer for sparse feature map generation. As pointed out in [50] and [51], decoupling $\mathbf{A}^{\mathrm{T}}$ from $\mathbf{A}$ allows for accelerated convergence and can be interpreted as learning a preconditioner for ISTA. Motivated by this, $\mathbf{A}^{\mathrm{T}}$ and $\mathbf{A}$ are implemented with two different networks. In this way, the sparse optimizer does not exactly fit the intermediate output of traditional ISTA and may exceed ISTA.

As presented in Fig. 4(a), the feature extractor $\mathbf{A}^{\mathrm{T}}$ is replaced with $\mathcal{E}(\cdot)$, which consists of a series of 3-D convolutional operations followed by the rectified linear unit (ReLU) nonlinear activation functions. Absorbing α, the feature decoder $\mathbf{A}^{\mathrm{T}}$ is implemented with $\mathcal{D}(\cdot)$, a combination of 3-D deconvolutional filters separated by ReLU. The sparsity inducer is a nonlinear function imposed on each entry of $\mathbf{s}$ and can be described in the vector form. It can be realized by $\mathrm{ReLU}(\mathbf{s} - \mathbf{b}) - \mathrm{ReLU}(-\mathbf{s} - \mathbf{b})$ where thresholding parameters $\mathbf{b}$ are associated with α and can also be learned from the data. The benefit of this realization is that it does not need to manually tune λ as in (3). Using $\mathcal{E}(\cdot)$ and $\mathcal{D}(\cdot)$, the sparse optimizer is formally defined by

$$\mathcal{S}_{k+1} = f_{\mathrm{soft}_{\mathbf{b}_k}}(\mathcal{S}_k + \mathcal{E}(\mathcal{Y} - \mathcal{D}(\mathcal{S}_k))), \quad k = 0, \ldots K \tag{7}$$

where $\mathcal{S}_k$ is the feature map in the kth optimizer. We untie the thresholding parameters $\mathbf{b}$ for DPNet-S, i.e., $\mathbf{b}$ is layerwise independent. As reported in [44], this setting is beneficial for image denoising. For DPNet-U, we use the same U-Net for all the layers because the untied setting results in more parameters to be learned. In practice, the initial feature map $\mathcal{S}_0$ can be set as zeros. As a comparison between (3) and (7), the sparse optimizer resembles ISTA but replaces the matrix computation with convolution/deconvolution and ReLU operations so that we can leverage the powerful representation ability of CNNs to learn a universal optimizer.

2) Image Reconstructor: $\mathbf{A}$ in (4) reconstructs the clean HSI based on the SR, i.e., $\widehat{\mathbf{s}}$. It can be seen from Fig. 4(b) that the image reconstructor $\mathcal{R}$ uses 3-D deconvolution filters followed by the ReLU activation function to map the extracted sparse feature map $\mathcal{S}_K$ to estimated HSI $\widehat{\mathcal{X}}$. In this way, the image reconstructor can be formally defined as

$$\widehat{\mathcal{X}} = \mathcal{R}(\mathcal{S}_{K+1}). \tag{8}$$

From Fig. 3, our DPNet-S is formed with multiple sparse optimizers, followed by an image reconstructor. All the key components of DPNet-S are induced from the SR-based HSI denoising model in (2)–(4), enabling our network to benefit from the classic SR model. Different from the original SR-based HSI denoising, our network replaces the matrix multiplication with convolutional and deconvolutional modules, which have stronger representation ability and can exploit additional structures beyond (2)–(4). For example, both convolutional and deconvolutional operators work locally but can exploit the global information at the same time due to shifting across the image. Moreover, the convolutional and deconvolutional operations also avoid extracting overlapped cubes and enlarge the receptive field to produce a more robust recovery of image structures. Discriminatively learning from data can also compensate for the model-mismatch problem of the SR model [11]. In a word, the hybrid advantages of the SR model and DL enable the high denoising ability of DPNet-S.

3) Extension to General Priors: As our network is based on the SR model rather than simply stacking many modules by trial, our network is quite general in nature, making further improvement easily deployable. For example, we can extend the denoising model in (2) by replacing the sparsity constraint with the general constraint $\mathcal{C}$ to account for the general priors, leading to

$$\min_{\mathbf{s}} \frac{1}{2} \left\| \mathbf{y} - \mathbf{A}\mathbf{s} \right\|_F^2 + \lambda \mathcal{C}(\mathbf{s}). \tag{9}$$

As we know, ISTA can be considered as a special case of the proximal gradient (PG) algorithm. Under the framework of the PG algorithm, the update rule accordingly becomes

$$\mathbf{s}_{k+1} = \mathrm{prox}_{\mathcal{C}}(\mathbf{s}_k + \alpha(\mathbf{A}^{\mathrm{T}}(\mathbf{y} - \mathbf{A}\mathbf{s}_k))) \tag{10}$$

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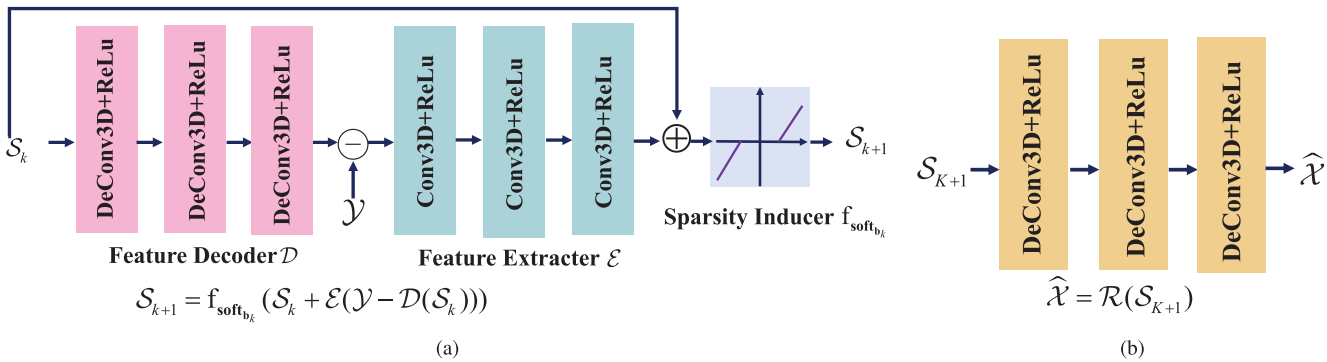


Fig. 4. Key components in the proposed DPNet-S. (a) Details of the sparse optimizer. (b) Details of the image reconstructor.

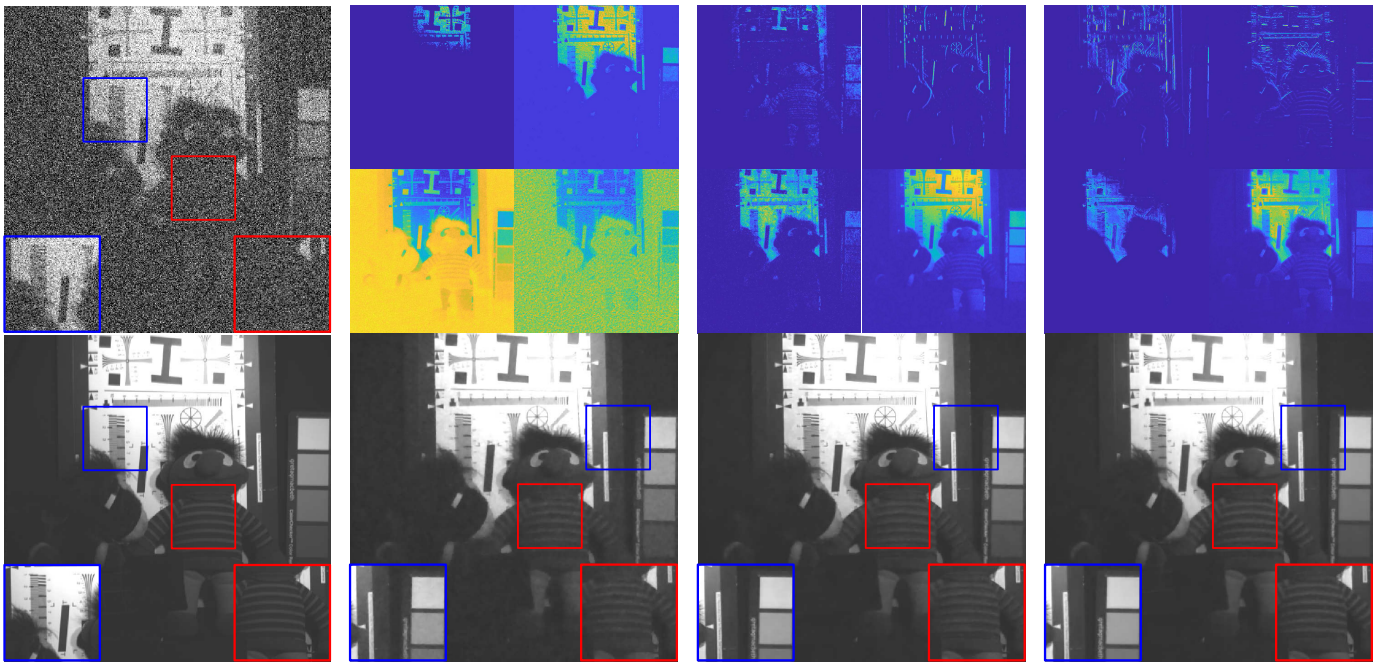


Fig. 5. Comparisons in terms of extracted feature maps and restored image. Because of space limitations, we only provide four typical feature maps (zoom in to see details). (a) Noisy&GT. (b) Unfolded network from (12). (c) DPNet-S. (d) DPNet-U.

where $\mathrm{prox}_{\mathcal{C}}$ is the proximal function for constraint $\mathcal{C}$. As revealed in [52], all the feedforward networks are universal approximations. Therefore, instead of the soft shrinkage operator, $\mathrm{prox}_{\mathcal{C}}$ can be implemented with different kinds of networks. U-Net [53] has powerful multiscale representation capability and proven denoising ability. To this end, we further replace the sparsity inducer with U-Net to more effectively capture the structure of $\mathbf{s}$ and denote the network as DPNet-U.

4) Connection With CSCNet: CSCNet [54] is obtained by unfolding the numerical optimization of the convolutional sparse coding (CSC) model and shares a similar architecture with our DPNet-S. When the ReLU activation function is removed, our DPNet-S can be considered to be obtained by unfolding the iterative optimization of the following overparameterized CSC model:

$$\min_{\mathcal{S}} \frac{1}{2}\left\|\mathcal{Y} - \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \mathcal{A}_3 \otimes \mathcal{S}\right\|_F^2 + \lambda\|\mathcal{S}\|_1 \tag{11}$$

where $\{\mathcal{A}_i\}_{i=1,2,3}$ are a set of convolutional dictionaries, $\mathcal{S}$ is the sparse feature map, and $\otimes$ denotes the convolutional operator.[1] Setting the number of convolutional dictionaries to one, the model becomes the traditional CSC model, i.e.,

$$\min_{\mathcal{S}} \frac{1}{2}\left\|\mathcal{Y} - \mathcal{A} \otimes \mathcal{S}\right\|_F^2 + \lambda\|\mathcal{S}\|_1. \tag{12}$$

As pointed out by Papyan et al. [55], to guarantee the quality of the CSC model, the similarity between atoms, i.e., the mutual coherence of the dictionary, should be small. However, the natural images usually consist of piecewise smooth regions and occasional textures. This property requires smooth or piecewise smooth filters, i.e., atoms, for SR, implying that the mutual coherence is unnecessarily small. This contradiction prevents the direct application of the CSC model to image modeling. For this reason, in super-resolution [56] and JPEG

[1]The iterative update rules of (11) and corresponding unfolded network architecture are provided in the Appendix.

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artifacts removal [57] tasks, the given image is generally separated into smooth and nonsmooth parts. The CSC model is applied only in the nonsmooth parts that contain rich details, thus avoiding the above conflict. To make the CSC model applicable to image denoising, CSCNet uses strided convolutions to generate multiple estimates for a given image and then averages these estimates to produce the denoised image.

In contrast to the CSCNet, we overparameterize the dictionary with multiple ones. Assuming that the dictionaries are known in advance, the model in (11) can be considered as the objective function of the following model:

$$\begin{aligned} \mathcal{Y} &= \mathcal{A}_1 \otimes \mathcal{S}_1 \\ \mathcal{S}_1 &= \mathcal{A}_2 \otimes \mathcal{S}_2 \\ \mathcal{S}_2 &= \mathcal{A}_3 \otimes \mathcal{S}, \quad \text{s.t. } \|\mathcal{S}\|_1 \leq K. \end{aligned} \tag{13}$$

Equation (13) provides a multilayer decomposition of a given HSI. More specifically, $\mathcal{A}_1$ contains local atoms to describe the image details. $\mathcal{A}_1 \otimes \mathcal{A}_2$ merges them to describe larger image structures that are more related to image edges. The further combination can widen the receptive field to capture more complex structures. By a multilayer composition of simpler structures, on the one hand, we have more ability to model complex relationships among the images. On the other hand, the deeper combination can provide atoms with lower mutual correlation [58], which is beneficial to ensure the quality of the SR. Moreover, the ReLU activation function also greatly contributes to the representativeness of the network [59]. Therefore, DPNet-S has a larger representation capacity and can achieve more effective HSI denoising. Accordingly, the DPNet-U can be considered as the network induced from the model in (14) that replaces the sparse prior with the general prior embodied by the U-Net

$$\min_{\mathcal{S}} \frac{1}{2}\left\|\mathcal{Y} - \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \mathcal{A}_3 \otimes \mathcal{S}\right\|_F^2 + \lambda \mathcal{C}(\mathcal{S}). \tag{14}$$

For more intuition, we provide four typical feature maps extracted by the unfolded network of (12), DPNet-S, and DPNet-U as well as the recovered band image in Fig. 5. As seen, the network obtained from (12) tends to extract the feature maps containing nonsmooth signals related to noises. In contrast, DPNet-S and DPNet-U prefer extracting the image textures that are beneficial to restore the image. Moreover, due to the powerful multiscale representation ability enabled by the U-Net, DPNet-U is more capable of extracting structural features and better recovering image details than DPNet-S.

C. Implementation Details

For both DPNet-S and DPNet-U, the feature extractor contains three convolutional layers. All the convolutional layers are of size $3 \times 3 \times 3$ with stride set 1, and the numbers of channels are set as 16, 32, and 64. The number of deconvolutional layers in both the feature decoder and image reconstructor is set as 3. The number of channels for both models gradually reduces from 64 and 32 to 16 and then to 1. K indexes the number of iterative steps in the sparsity optimizer and is set to 4, considering the training efficiency. The 3-D residual U-Net involving three scales is used in the DPNet-U denoiser. The influence of these configurations on the denoising performance is analyzed in Section IV.

Given a set of training samples $\{\mathcal{Y}_i, \mathcal{X}_i\}_{i=1}^N$, a mean square error is adopted as the loss function for both methods, i.e.,

$$\mathcal{L}(\Theta) = \frac{1}{N}\sum_{i=1}^{N} \left\| f(\Theta, \mathcal{Y}_i) - \mathcal{X}_i \right\|_F^2 \tag{15}$$

where Θ is the parameter set to be learned.

IV. Experiments

In this section, we present the experimental setting, network analysis, and results.

A. Experimental Setting

1) Training Datasets: Following [31] and [42], we selected 100 HSIs from the ICVL HSI dataset[2] to generate noisy–clean HSI pairs for training. Subimages' size of $64 \times 64 \times 31$ was extracted from the original image containing 1392×1300 pixels and 31 bands to train the network. Data augmentation, including random flipping, cropping, and resizing, was performed to enrich the training set.

2) Training Details: DPNet-S and DPNet-U were implemented on the PyTorch platform and trained using two NVIDIA GeForce RTX 3090 GPUs through the Adam optimizer. The batch size was set as 8. The initial learning rate was assigned as 5×10^{-4} and then cut to half in every ten epochs. It took 50 epochs for network convergence.

3) Compared Methods: Our methods were compared with 12 methods in the literature, including seven model-based methods, i.e., BM4D [60], LRMR [12], MTSNMF [6], LLRT [61], NGMeet [14], 3DLogTNN [21], and FastHyDe [13]; and five DL-based methods, i.e., CSCNet [54], SMDS-Net [42], TRQ3D [35], MAC-Net [42], and QRNN3D [31]. All the DL-based methods were retrained, following the setting in the original paper. We tried our best to tune the parameters of model-based methods to the optimal.

4) Evaluation Indexes: Peak signal-to-noise ratio (PSNR), structure similarity (SSIM), and spectral angle mapper (SAM) are used to evaluate the denoising performance. PSNR and SSIM measure the quality of reconstruction of lossy compression and perceived changes in structural information. SAM depicts the spectral differences between clean and denoised HSIs. Smaller SAM and larger PSNR and SSIM imply better HSI denoising.

5) Noise Generation: Two types of noise are added to produce the simulated data, i.e., non-i.i.d. Gaussian noise and mixture noise. The detailed noise configurations are given as follows.

1) *Non-i.i.d. Gaussian Noise:* As mentioned earlier, the noise intensities among bands are always different in real-world cases. Therefore, it is always expected that a denoising method can effectively handle the noise differences among bands so as to meet the requirement of practical HSI denoising. Following [6], [13], and [21],

[2]Link: http://icvl.cs.bgu.ac.il/hyperspectral/

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TABLE I
IMPACT OF RELU ON DPNET-S

Index	With	Without
PSNR	**42.95**	41.51
SSIM	**.9772**	.9694
SAM	**.0574**	.0725

different intensities of zero-mean Gaussian noise were added to each band to produce the noisy HSIs. The noise levels were assigned in three continuous ranges, i.e., [0, 15], [0, 55], and [0, 95].

2) *Mixture Noise:* Remote sensing images are usually contaminated with mixture noises. Therefore, following [31], the mixture noise is generated with a combination of non-i.i.d. Gaussian noise, impulse noise, strips, and deadlines. The noise level of Gaussian noise was set in the range [0, 95]. One third of the bands were contaminated by the impulse noise with intensities between 0.1 and 0.7. One third of the bands were contaminated by the strip noise with columns of 5%–15%. One third of the bands were chosen to be degraded by the deadline noise.

B. Network Analysis

Here, we conducted a study to show the connection between CSCNet and the impact of training samples and key parameters. All experiments were conducted on the ICVL dataset with $\sigma \in [0,\ 95]$.

1) Comparison With CSCNet: Here, we compare our approach with CSCNet, which was originally designed for color and gray images and is not directly applicable to HSIs. For this reason, we slightly modified the original 2-D CSCNet to a 3-D CSCNet to handle HSIs more efficiently by changing the 2-D convolution to the 3-D convolution and setting the stride in the spectral domain as 1. The other settings remained the same as the original implementation, i.e., the spatial kernel size and stride respective were set as 11 and 7, respectively.

Setting the number of layers the same, we first compare our DPNet-S with CSCNet in Fig. 6. In the original implementation of CSCNet, the feature dimension is set to 175, which is too computationally intensive for our DPNet-S. For this reason, we omitted this setting for DPNet-S. Extracting more features helps to increase the representation ability of the network. As a result, the denoising performance of both CSCNet and DPNet-S improves. With the same dimensionality, our DPNet-S greatly outperforms CSCNet. Furthermore, even when setting the feature dimensionality of CSCNet about twice that of DPNet-S, DPNet-S still dominates CSCNet. The possible reason is that CSCNet still needs patch averaging to generate the denoised HSIs, which is not the case for our DPNet-S. In addition, the combination of multiple convolutions also helps DPNet-S more effectively extract features from simple to complex for improved denoising.

2) Impact of ReLU on DPNet-S: As mentioned above, when the "ReLU" activation function is removed, DPNet-S can be regarded as an unfolding network obtained from (13). Table I presents a study of the effect of the ReLU activation function. With the help of the ReLU, the network representation can be further enhanced to achieve higher performance.

TABLE II
ABLATION STUDY ON THE SPARSITY INDUCER

w/o	PSNR	SSIM	SAM	#Params
With	**42.79**	**.9745**	**.0647**	208912
Without	40.85	.9622	.0794	**208656**

TABLE III
IMPACT OF K ON DENOISING PERFORMANCE. "S" AND "U," RESPECTIVELY, STAND FOR DPNET-S AND DPNET-U

K	2		3		4		5	
	S	**U**	**S**	**U**	**S**	**U**	**S**	**U**
PSNR	40.89	44.31	42.16	44.72	42.95	44.72	43.26	44.97
SSIM	.9638	.9812	.9638	.9831	.9772	.9827	.9774	.9835
SAM	.0762	.0466	.0762	.0445	.0574	.0451	.0553	.0431

TABLE IV
IMPACT OF U-NET ON DENOISING PERFORMANCE

#Layer	1	2	3	4	5	6
PSNR	43.87	44.10	44.40	44.72	44.97	44.97
SSIM	.9805	.9813	.9820	.9827	.9839	.9839
SAM	.0502	.0511	.0481	.0451	.0429	.0427

3) Ablation Study on the Sparsity Inducer of DPNet-S: We conduct an ablation study on the sparsity inducer to see its impact on HSI denoising. It can be seen from Table II that the sparsity inducer module greatly improves the HSI denoising performance despite increasing very few parameters. The main reason is that SR can effectively capture the low-dimensional structure of the underlying clean HSI.

4) Influence of the Number of Optimizers: K indexes the number of optimizers for feature learning. Table III shows that both DPNet-S and DPNet-U gradually converge when K increases from 2 to 5. On the other hand, we found that the training becomes difficult and the loss suddenly increases during training, as the number of layers increases. The possible reason is that the feature decoder and extractor are shared among the layers, making the learning strongly coupled as the number of layers increases. As the larger values of K also imply more parameters, we suggest setting $K = 4$, which can well balance two factors. Fig. 7 visualizes the feature extracted in each optimizer. As optimization progresses, feature maps show clearer structures. This is in line with the traditional optimization method, i.e., with several iterations, the learned representation will reach convergence.

5) Impact of U-Net on DPNet-U: Here, we investigate the impact of the scale of U-Net on the denoising performance of DPNet-U. Table IV shows the changes in denoising performance when the number of encoding layers is set from 1 to 6. In general, more encoding layers facilitate exploring the multiscale structure of HSIs, thus achieving higher denoising performance. As more layers also imply more parameters to be learned, we set the number of encoding layers as 5, corresponding to three scales.

6) Impact of Training Samples: A benefit of integrating the model into DL-based methods is that we can alleviate the requirement for large-scale training samples. Table V shows the changes in denoising performance when setting the number of training samples from 10, 20, 40, and 80 to 100. Overall, more training samples contribute to better denoising. However, we notice that the PSNR index of DPNet-S and DPNet-U only increases by 0.79 and 1.01 dB, respectively,

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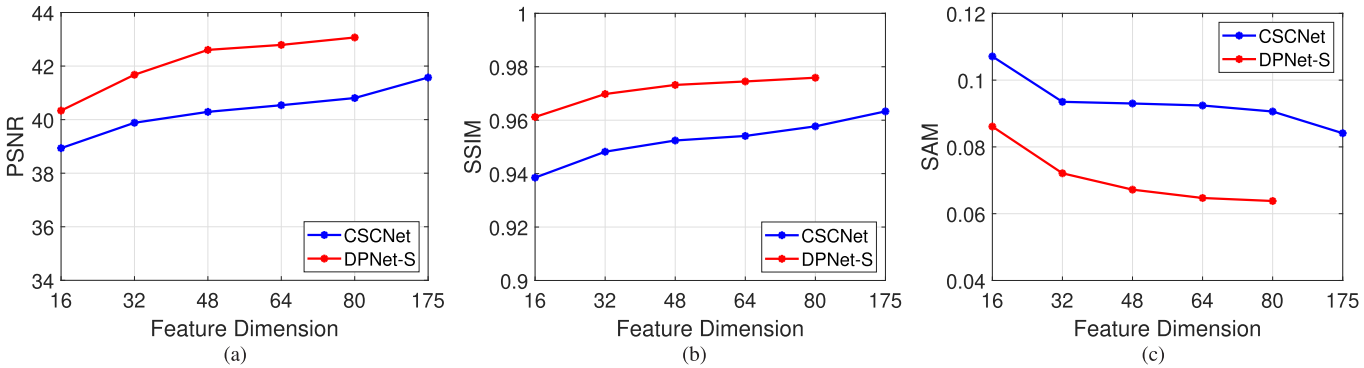


Fig. 6. Comparison with CSCNet regarding different feature dimensionalities. (a) PSNR. (b) SSIM. (c) SAM.

TABLE V

IMPACT OF THE NUMBER OF TRAINING SAMPLES. "S" AND "U" STAND FOR DPNET-S AND DPNET-U, RESPECTIVELY

#Samples	10		20		40		80		100	
	S	**U**	**S**	**U**	**S**	**U**	**S**	**U**	**S**	**U**
PSNR	39.89	42.99	42.16	43.71	42.74	44.40	42.92	44.75	42.95	44.72
SSIM	.9559	.9776	.9736	.9801	.9763	.9820	.9761	.9830	.9772	.9827
SAM	.0843	.0542	.0615	.0501	.0589	.0457	.0589	.0453	.0574	.0451

TABLE VI

COMPARISON OF DIFFERENT METHODS ON 50 TESTING HSIS OF ICVL DATASET. THE TOP TWO VALUES ARE MARKED RED AND BLUE

			Model-based methods							**DL-based methods**						
σ	Index	Noisy	BM 4D	LRMR	MTS NMF	3DLog TNN	LLRT	NG Meet	Fast HyDe	CSC Net	SMDS-Net	TRQ 3D	QRN N3D	MAC -Net	**DP Net-S**	**DP Net-U**
[0-15]	PSNR	33.18	44.39	41.50	45.39	43.00	45.74	39.63	48.08	49.04	47.20	46.43	47.84	48.21	**50.24**	**51.25**
	SSIM	.6168	.9683	.9356	.9592	.9872	.9657	.8612	.9917	.9911	.9871	.9878	.9913	.9915	**.9934**	**.9944**
	SAM	.3368	.0692	.1289	.0845	.0442	.0832	.2144	.0404	.0358	.0444	.0437	.0373	.0387	**.0299**	**.0251**
[0-55]	PSNR	21.72	37.63	31.50	38.02	32.29	36.80	31.53	42.86	43.97	42.23	44.64	44.33	43.74	**44.82**	**46.55**
	SSIM	.2339	.9008	.6233	.8586	.6542	.8285	.6785	.9800	.9770	.9668	.9840	.9809	.9768	**.9832**	**.9874**
	SAM	.7012	.1397	.3583	.2340	.4993	.2316	.4787	.0630	.0583	.0675	.0487	.0553	.0582	**.0480**	**.0367**
[0-95]	PSNR	17.43	34.71	27.00	34.81	24.28	31.89	27.62	40.84	41.78	39.60	**43.54**	41.08	41.24	42.95	**44.72**
	SSIM	.1540	.8402	.4208	.7997	.4200	.6885	.5363	.9734	.9663	.9496	**.9806**	.9541	.9577	.9772	**.9827**
	SAM	.8893	.1906	.5142	.3266	.7985	.3444	.6420	.0771	.0725	.0841	**.0523**	.0920	.0841	.0574	**.0451**
Mixture	PSNR	13.21	23.36	23.10	27.55	17.70	18.23	23.61	27.58	37.63	27.71	**39.73**	38.43	28.44	38.61	**40.91**
	SSIM	.0841	.4275	.3463	.6743	.2408	.1731	.4448	.7250	.9208	.8078	**.9491**	.9302	.7393	.9312	**.9489**
	SAM	.9124	.5476	.5144	.5326	.8368	.6873	.6252	.4534	.1473	.4052	**.0869**	.1154	.4154	.1178	**.0971**

when training samples increase from 20 to 100. This means that our networks do not heavily rely on large-scale training samples to learn the parameters, showing their strong learning ability. We attribute this phenomenon to the physical models that guide the network to utilize the structure of the denoising problem and the physical properties of HSIs more effectively.

C. Comparison With State-of-the-Art Methods

1) Comparison on ICVL Dataset: We first report the experimental results on the ICVL dataset in Table VI, which contains 50 testing images. Similar to [31], the images that contain 512×512 pixels and 31 bands were cropped from the original image for testing. For the CSCNet, the feature dimension was 175, as in the original implementation. In the non-i.i.d. Gaussian noise cases, our DPNet-U produces the overall best denoising results in all the evaluation criteria and noise ranges, confirming its effectiveness and robustness for HSI denoising. NGMeet assumes all bands to follow the same noise distribution when estimating the spectral basis. Consequently, when nonidentically distributed noises are present, the estimated spectral basis becomes inaccurate, leading to poor denoising performance. Except for 3DLogTNN and LRMR, all other model-based methods conduct HSI denoising by averaging the denoising results of overlapping cubes or patches. Compared with them, our DPNet-S and DPNet-U provide significantly better performance because of performing denoising on the global HSIs directly and the hybrid

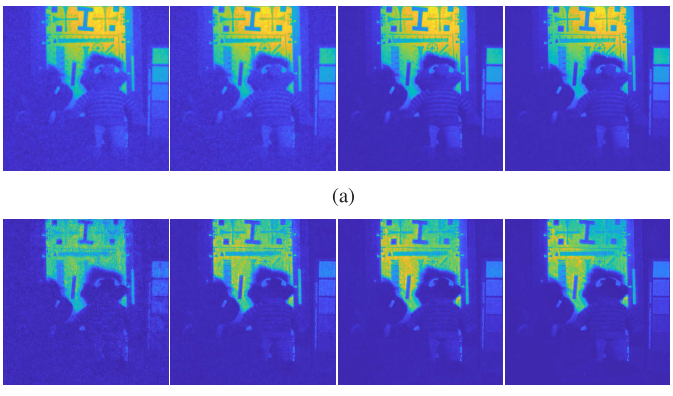


Fig. 7. From left to right, the rows are feature maps extracted in each layer when $K = 4$ (zoom in to see details). (a) DPNet-S. (b) DPNet-U.

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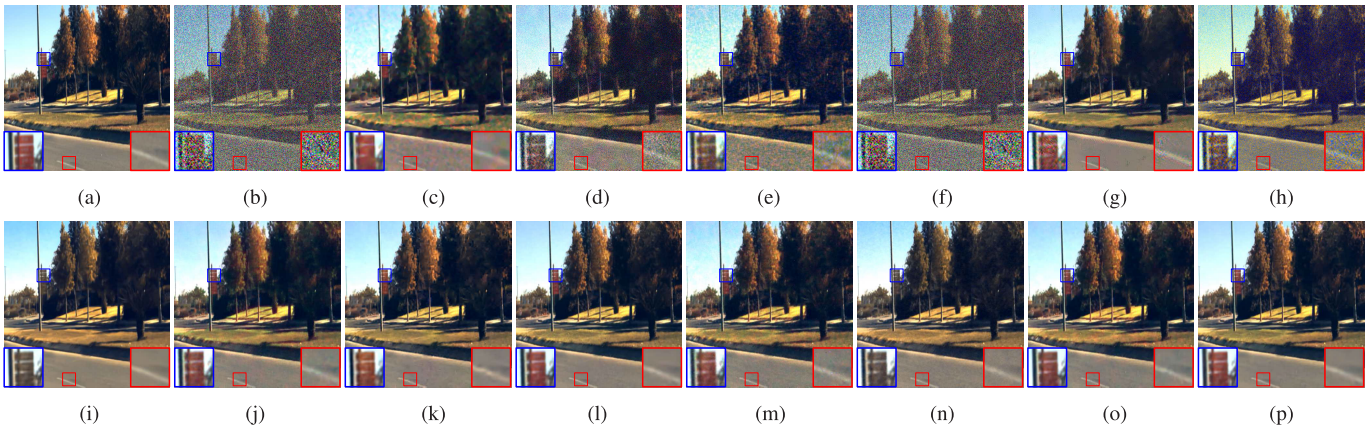


Fig. 8. Denoising results on the lehavim_0910-1605 image in the ICVL dataset with the noise variance in [0, 95]. The false-color images were generated by combining bands 31, 21, and 9 (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

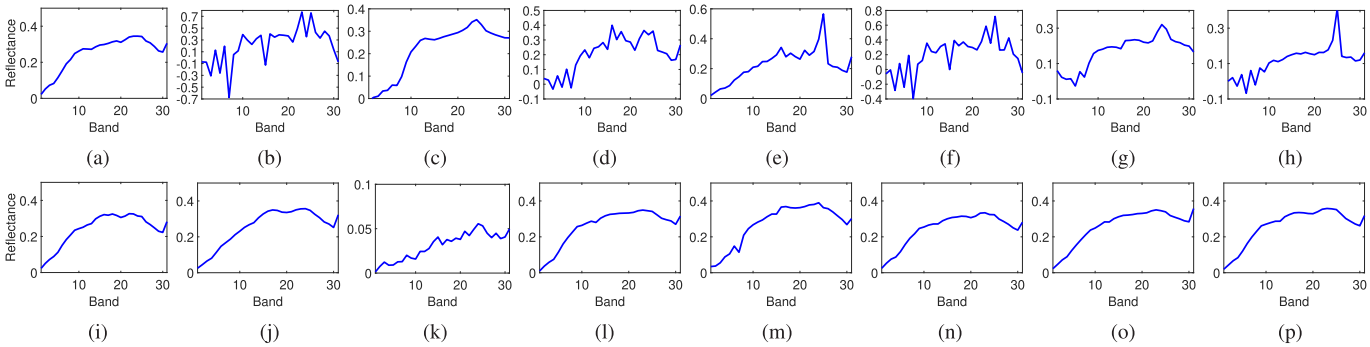


Fig. 9. Spectrum of pixel (225, 268) in lehavim_0910-1605 image with the noise variance in [0, 95] (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

TABLE VII

COMPARISON OF DIFFERENT METHODS ON 32 TESTING HSIS OF CAVE DATASET. THE TOP TWO VALUES ARE MARKED RED AND BLUE

			Model-based methods							DL-based methods						
σ	Index	Noisy	BM 4D	LRMR	MTS NMF	3DLog TNN	LLRT	NG Meet	Fast HyDe	CSC Net	SMDS-Net	TRQ 3D	QRN N3D	MAC-Net	**DP Net-S**	**DP Net-U**
[0-15]	PSNR	33.37	44.76	39.53	42.82	39.24	45.02	37.81	42.15	46.25	45.40	40.77	41.52	44.50	**47.27**	**48.42**
	SSIM	.6236	.9742	.9417	.9691	.9765	.9634	.8628	.9751	.9874	.9826	.9766	.9771	.9839	**.9902**	**.9921**
	SAM	.6150	.1297	.2582	.1361	.1174	.1773	.4226	.1759	.0901	.1222	.1340	.1377	.1222	**.0841**	**.0677**
[0-55]	PSNR	21.51	38.04	33.47	35.63	29.39	36.34	28.22	36.57	41.48	39.28	37.90	38.84	38.06	**41.65**	**42.39**
	SSIM	.2389	.9102	.7629	.8413	.6022	.8244	.5788	.9292	.9691	.9510	.9621	.9530	.9339	**.9746**	**.9788**
	SAM	1.001	.2936	.5383	.3764	.8401	.4446	.8359	.3417	.1474	.1671	.1770	.2074	.2704	**.1393**	**.1249**
[0-95]	PSNR	17.55	34.93	30.40	32.57	22.83	31.59	24.47	35.44	39.28	36.22	36.79	36.47	35.58	**39.57**	**40.40**
	SSIM	.1632	.8330	.6166	.7677	.4092	.6868	.4367	.9073	.9540	.9247	.9476	.8994	.8798	**.9624**	**.9699**
	SAM	1.141	.4305	.7205	.5191	1.082	.6103	.9619	.3683	.1796	.1977	.2269	.3065	.3751	**.1694**	**.1483**
Mixture	PSNR	13.01	22.69	22.86	25.82	17.50	18.12	21.16	25.18	35.07	26.84	34.36	33.99	25.56	**35.08**	**36.05**
	SSIM	.0939	.4037	.3913	.6327	.2405	.1881	.3922	.6398	.8668	.7115	**.8730**	.8445	.5582	.8533	**.8779**
	SAM	1.100	.8004	.7848	7690	1.050	.9047	.8626	.7211	.3238	.6701	.3189	.3402	.7045	**.3157**	**.2853**

advantages of SR model and data-driven training. Compared with other DL-based methods, the embedded physical models better capture the low-dimensional subspace of clean HSIs with fewer parameters. As a result, our DPNet-S and DPNet-U dominate other DL-based methods to a certain margin. In addition, DPNet-U obtains higher denoising effectiveness than DPNet-S as DPNet-U employs U-Net to learn general deep priors, which are more robust than handcrafted priors. In the mixed noise case, the heavy noise makes it difficult to select optimal parameters for effective denoising. Therefore, poor performance is provided. SMDS-Net and MAC-Net assume that the noises follow the Gaussian distribution to build the network and produce inferior performance. It is worth noting that our DPNet-S and DPNet-U achieve the overall best performance.

We further compare the denoised HSIs produced by all methods on the lehavim_0910-1605 image with the noise variance in [0, 95]. As shown in Fig. 8, among model-based methods, FastHyDe obtains the best visually favorable results due to the consideration of global spectral low rankness and nonlocal spatial sparsity. In contrast to alternative DL-based methods, DPNet-S and DPNet-U can remove noises with

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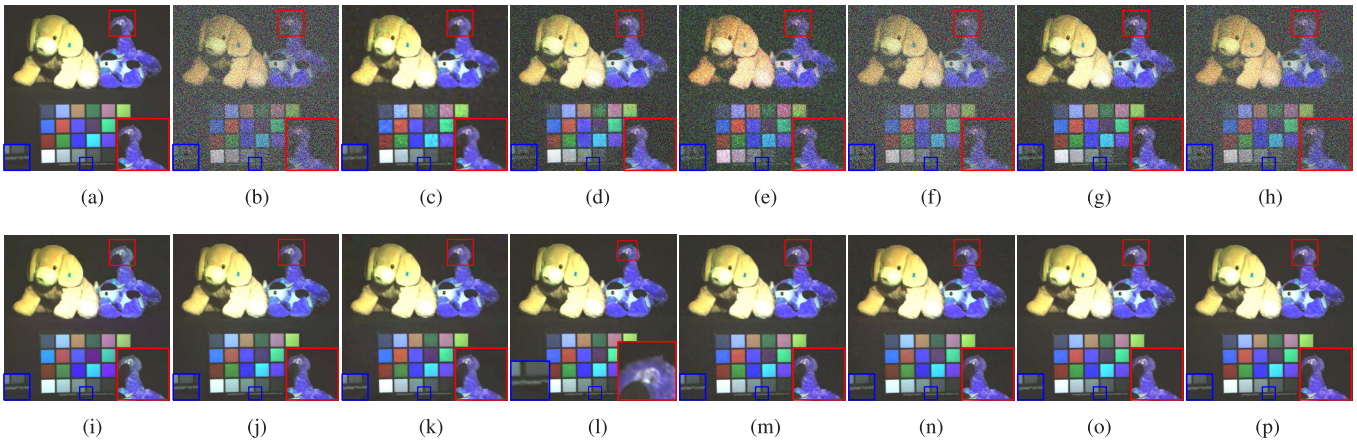


Fig. 10. Denoising results on the stuffed toys HSI in the CAVE dataset with the noise variance in [0, 95]. The false-color images were generated by combining bands 7, 15, and 25 (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

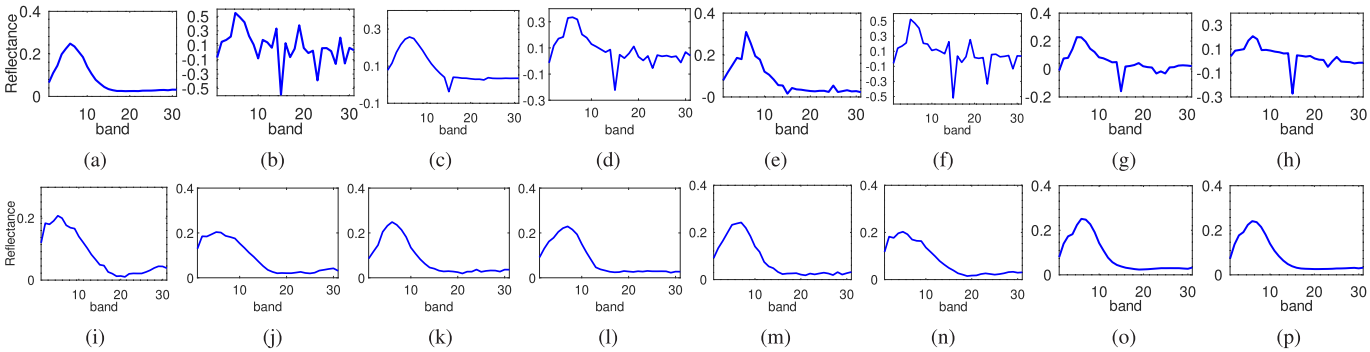


Fig. 11. Spectrum of pixel (130, 433) in stuffed toys image before and after denoising (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

TABLE VIII

COMPARISON OF DIFFERENT METHODS ON HOUSTON 2018 HSI. THE TOP TWO VALUES ARE MARKED RED AND BLUE

Index	Noisy	Model-based methods							DL-based methods						
		BM 4D	LRMR	MTS NMF	3DLog TNN	LLRT	NG Meet	Fast HyDe	CSC Net	SMDS Net	TRQ 3D	QRN N3D	MAC-Net	**DP Net-S**	**DP Net-U**
PSNR	11.72	22.76	21.84	25.86	15.17	15.58	22.36	27.07	32.62	24.99	32.55	31.88	28.93	**33.92**	**34.26**
SSIM	.0843	.4762	.3914	.6933	.1876	.1386	.5169	.7757	.8891	.8729	.9194	.8749	.8080	**.9244**	**.9360**
SAM	.9778	.5168	.4857	.4977	.9106	.7652	.5728	.4518	.1181	.3878	.1241	.1472	.2537	**.1012**	**.1009**
Time (s)	-	508.3	22.9	86.4	422.0	2489.4	667.9	9.8	155.2	728.6	79.81	81.2	7.6	361.6	909.9

preserved predominant structures. DPNet-S, DPNet-U, and FastHyDe provide comparable denoised HSIs. Fig. 9 further plots the spectral reflectance of pixel (225, 268) before and after denoising. It can be observed that our DPNet-S and DPNet-U can also more precisely recover the spectrum. In summary, this experiment evidently confirms the denoising capability of our deep parameterized neural networks over the model and DL-based methods.

2) Comparison on CAVE Dataset: We further ran all the denoisers on the CAVE dataset, which includes 32 HSIs. In contrast to the ICVL dataset, which contains mostly outdoor scenarios, the majority of the HSIs in the CAVE dataset are captured indoor. Rather than fine-tuning the deep models, we directly used the models trained on the ICVL dataset. As seen from Table VII, without the guidance of the physical model, QRNN3D drops dramatically, even inferior to the model-based methods. Due to the preferable generalization ability of the physical model, other DL-based approaches achieve better performance. It is worth noting that our DPNet-S and DPNet-U outperform other methods because of the absence of the patch division process. Fig. 10 visualizes the denoised HSIs of all the methods by composing bands 7, 15, and 25 with $\sigma \in [0, 95]$. Alternative methods cannot remove all the noises, e.g., BM4D, MTSNMF, LLRT, and NGMeet. QRNN3D introduces additional structures. In contrast, the proposed DPNet-S and DPNet-U can best recover underlying clean HSI by best matching with the ground-truth HSI. Fig. 11 provides the recovered spectrum by all the methods. The results show that the proposed methods have substantially higher denoising ability.

3) Comparison on Houston 2018 HSI: We further ran all the methods on the Houston 2018 HSI. This is a remote sensing image captured with an ITRES CASI 1500 hyperspectral imager. There are 1202 $\times$ 4172 pixels and 48 bands ranging from 380 to 1050 nm. The first two bands are removed because of noise. The center part of the image with 512 $\times$ 512 pixels was used for the experiment. Considering that remote sensing HSIs were generally polluted by mixture noise,

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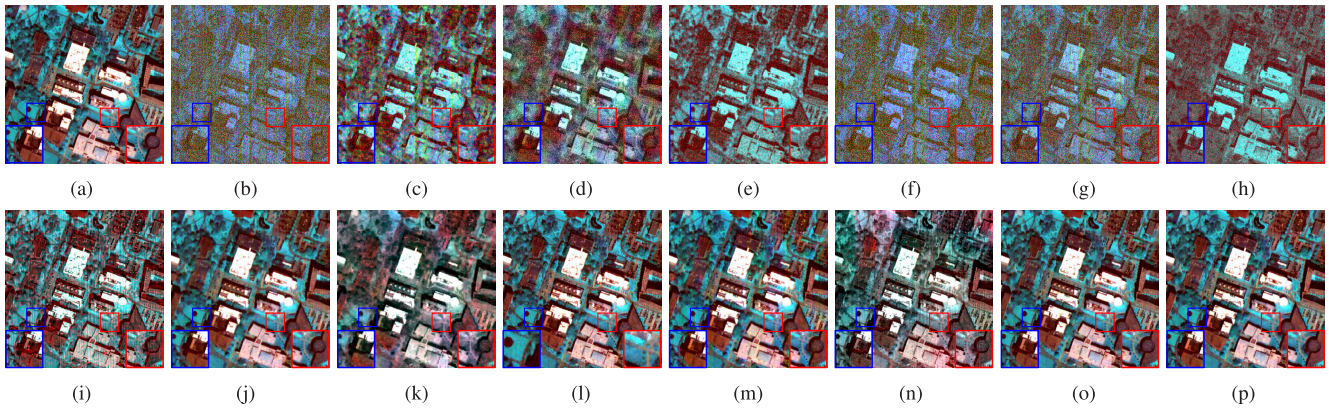


Fig. 12. Denoising results on Houston 2018 HSI in the case of mixture noises. The false-color images were generated by combining bands 4, 32, and 46 (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

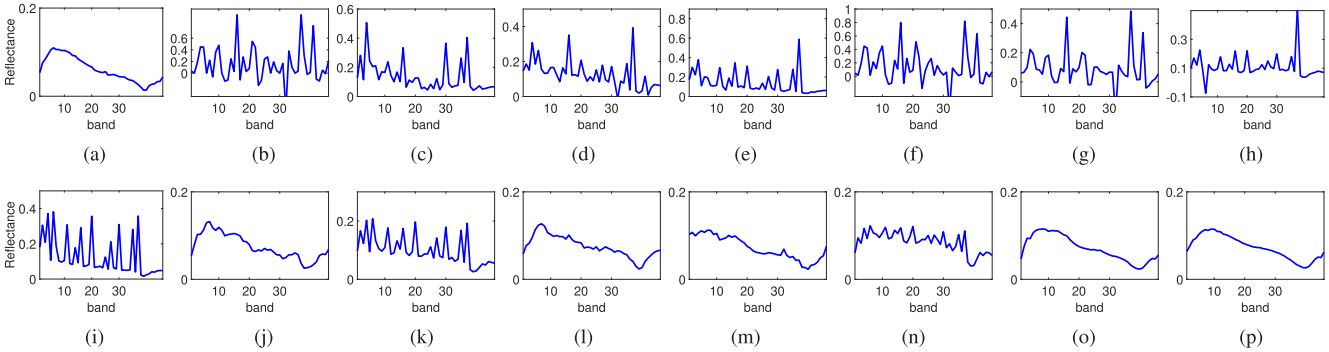


Fig. 13. Spectrum of pixel (208, 245) in Houston 2018 HSI in the case of mixture noises (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

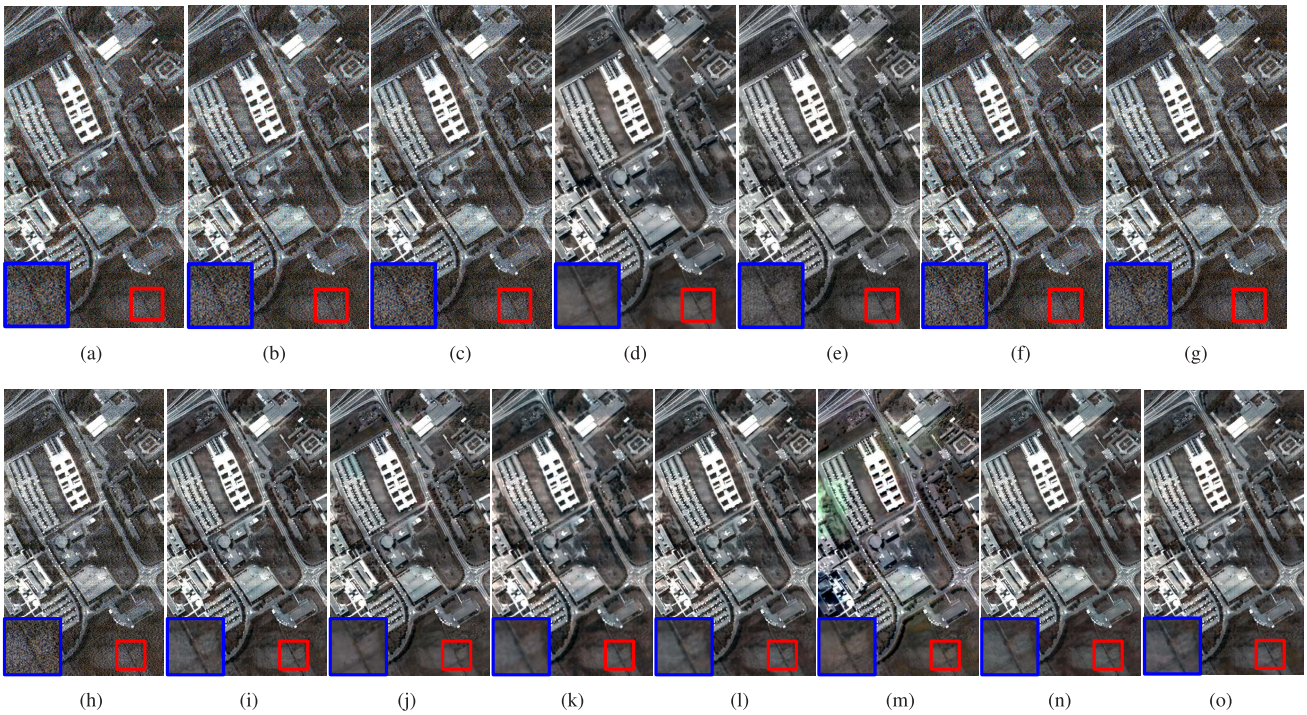


Fig. 14. Denoising results on the real-world Pavia University HSI. The false-color images were generated by combining bands 1–3 (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

we only compare all the methods in the mixture noise case in Table VIII. From Table VIII, our DPNet-S and DPNet-U also provide the most promising denoising among all the methods. We also provide the CPU time of all the methods.

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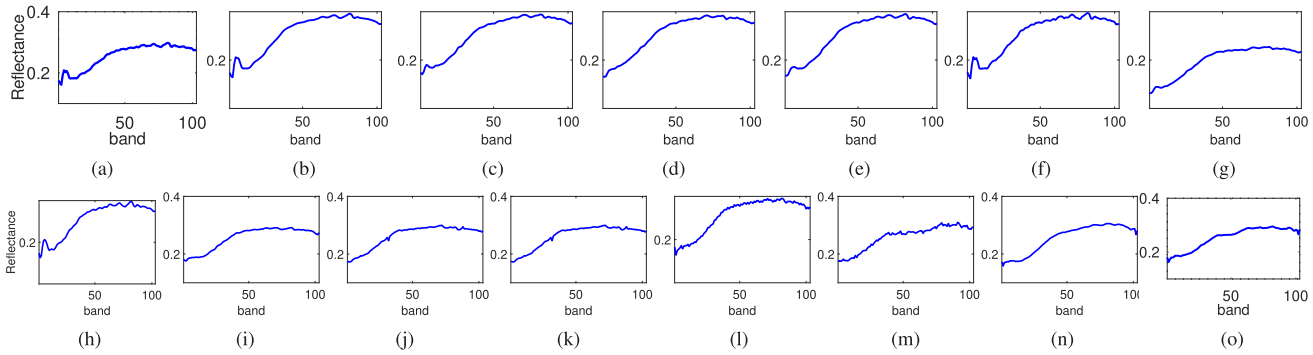


Fig. 15. Spectrum of pixel (160, 416) in the real-world Pavia University HSI (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

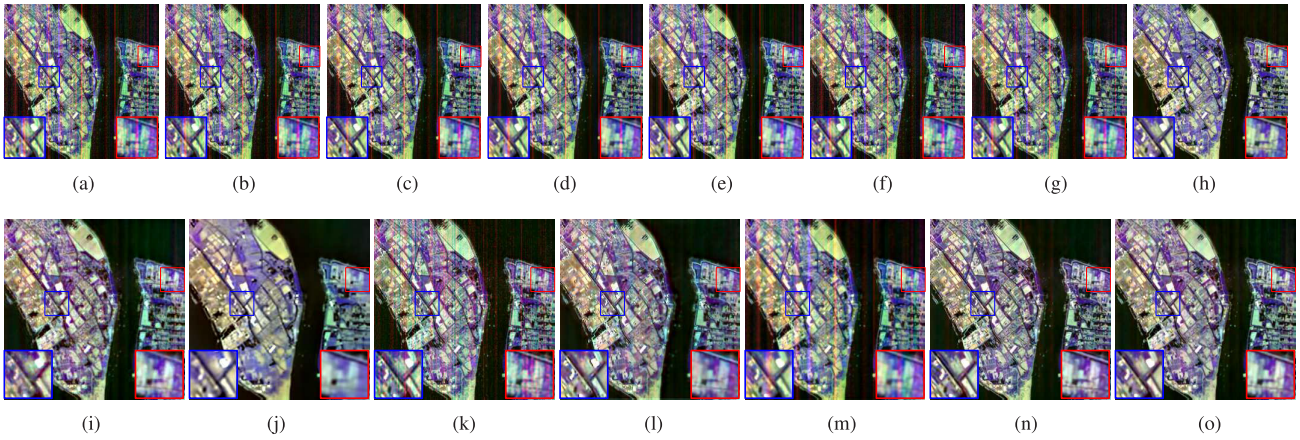


Fig. 16. Denoising results on GF-5 Shanghai HSI. The false-color images were generated by combining bands 154, 95, and 42 (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

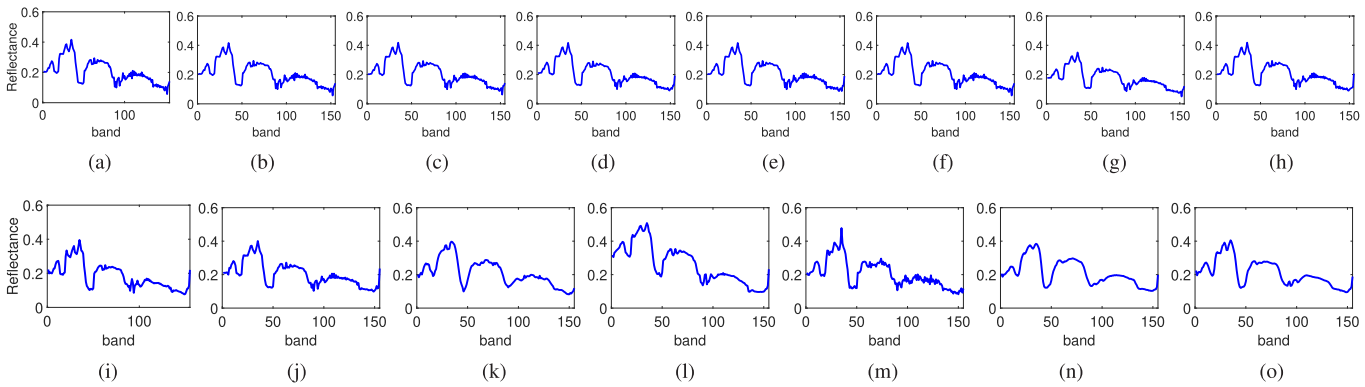


Fig. 17. Spectrum of pixel (65, 65) in GF-5 Shanghai HSI (zoom in to see details). (a) Clean. (b) Noisy. (c) BM4D. (d) LRMR. (e) MTSNMF. (f) 3DLogTNN. (g) LLRT. (h) NGMeet. (i) FastHyDe. (j) CSCNet. (k) SMDS-Net. (l) TRQ3D. (m) QRNN3D. (n) MAC-Net. (o) DPNet-S. (p) DPNet-U.

All the experiments are conducted on a machine with two Intel Xeon Silver 4114 CPUs at 2.20 GHz and 128-GB RAM. Our DPNet-S and DPNet-U occupy the most time among all the methods. Due to the GPU and parallel computing, the time can be greatly shortened. Moreover, as denoising is often performed offline, it is generally worth paying more time for performance improvement. We further give a comparison of all the methods with respect to recovered band images and spectrum in Figs. 12 and 13, respectively. Our DPNet-S and DPNet-U generally produce more satisfactory performance by best matching with the ground truth.

4) Comparison on Real-World Pavia University HSI: Pavia University HSI is a real-world remote sensing HSI with a higher spectral resolution, i.e., more spectral bands but lower spatial resolution than CAVE and ICVL HSIs. Specifically, the Pavia University HSI covers 610 × 340 pixels with 1.3-m spatial resolution and 103 bands. We also used the models trained on the ICVL dataset to test DL-based methods. Noises contaminate the first three bands. Because of no ground-truth HSI, we qualitatively compare all the methods. Figs. 14 and 15, respectively, show the recovered spatial images and spectral reflectance. There are

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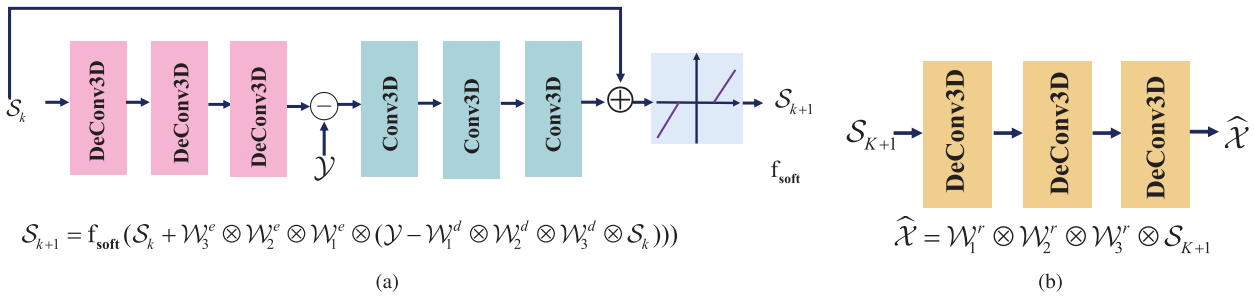


Fig. 18. Unfolding network of (11). (a) Generalized convolutional LISTA. (b) Image reconstruction.

residual noises in the results of BM4D, LRMR, 3DLogTNN, LLRT, NGMeet, FastHyDe, and CSCNet. The possible reason is that they use the "overlap-averaging" procedure for denoising. Comparatively, MTSNMF, DPNet-S, and DPNet-U can more competently remove the noises with preserved edges. i

5) Comparison on Real-World GF-5 Shanghai HSI: We further ran all the methods on the real-world Gaofeng-5 (GF-5) Shanghai HSI. The image was captured by the Advanced Hyperspectral Imager (AHSI) and cropped into a 300×300 size with 155 bands. To evaluate the DL-based methods, we directly applied the model trained on the ICVL dataset. As can be seen from Fig. 16(a), the GF-5 Shanghai HSI dataset was contaminated with mixed noises, including strips and deadlines. DL-based methods show promising denoising results. However, SMDS-Net and CSCNet tend to produce oversmoothed images, while TRQ3D and MAC-Net still exhibit noticeable residual noises. It is worth mentioning that our proposed DPNet-S and DPNet-U are more effective in removing the mixed noises. To gain further insight, Fig. 17 shows the recovered spectrum at pixel (65, 65). It can be observed that our DPNet-S and DPNet-U generate more accurate spectra compared to other methods, showing their superiority in HSI denoising.

V. Conclusion

This article incorporates deep architectures into the iterative optimization scheme of SR, yielding the DPNet-S denoiser. The two key components of DPNet-S, i.e., sparse optimizer and image reconstructor, can be seamlessly linked with ISTA-based SR and image reconstruction but overcome the limitations of SR-based HSI denoising in overlapping cube extraction and manually designed optimization. Moreover, we replace the sparsity inducer in DPNet-S with a U-Net to implicitly learn general priors from data. Thorough experiments and comprehensive analysis confirm the strong denoising ability of our methods. We will employ deep neural networks to parameterize a combination of multiple priors for improved HSI denoising performance in our future work.

Appendix
Unfolding Network of (11)

Here, we induce the update rule and the corresponding unfolded neural network of (11) in the main text. Assuming that the dictionaries are known, under the framework of the iterative soft thresholding algorithm, we can get the following update rule:

$$\begin{aligned}\mathcal{S}_{k+1} &= f_{\text{soft}}\Big(\mathcal{S}_k + \frac{1}{L}\text{flip}(\mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \mathcal{A}_3) \\ &\qquad \otimes (\mathcal{Y} - \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \mathcal{A}_3 \otimes \mathcal{S}_k)\Big) \\ &= f_{\text{soft}}\Big(\mathcal{S}_k + \frac{1}{L}\text{flip}(\mathcal{A}_3) \otimes \text{flip}(\mathcal{A}_2) \otimes \text{flip}(\mathcal{A}_1) \\ &\qquad \otimes (\mathcal{Y} - \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \mathcal{A}_3 \otimes \mathcal{S}_k)\Big) \quad (16)\end{aligned}$$

where $(1/L)$ is the step size and f_{soft} is a soft shrinkage operator defined as $\text{sign}(s)\max(|s|-b, 0)$ with the thresholding parameter b associated with hyperparameter λ. The operation $\text{flip}(\cdot)$ reverses the order of each atom of a dictionary in all dimensions.

Introducing two sets of fully trainable and independent variables $\{\mathcal{W}_1^e, \mathcal{W}_2^e, \mathcal{W}_3^e\}$ and $\{\mathcal{W}_1^d, \mathcal{W}_2^d, \mathcal{W}_3^d\}$, (11) can be unfolded into the following generalized convolutional LISTA structure:

$$\begin{aligned}&\mathcal{S}_{k+1} \\ &= f_{\text{soft}}\big(\mathcal{S}_k + \mathcal{W}_3^e \otimes \mathcal{W}_2^e \otimes \mathcal{W}_1^e \otimes \big(\mathcal{Y} - \mathcal{W}_1^d \otimes \mathcal{W}_2^d \otimes \mathcal{W}_3^d \otimes \mathcal{S}_k\big)\big). \quad (17)\end{aligned}$$

Note that we have absorbed the step size into $\{\mathcal{W}_1^e, \mathcal{W}_2^e, \mathcal{W}_3^e\}$. In general, $\{\mathcal{W}_1^d, \mathcal{W}_2^d, \mathcal{W}_3^d\}$ map the extracted sparse feature maps into the same space with $\mathcal{Y}$ and can be implemented with a set of deconvolutional operations. $\{\mathcal{W}_1^e, \mathcal{W}_2^e, \mathcal{W}_3^e\}$ extract the feature map from $\mathcal{Y} - \mathcal{W}_1^d \otimes \mathcal{W}_2^d \otimes \mathcal{W}_3^d \otimes \mathcal{S}_k$ and can be implemented with a set of convolutional operations. After obtaining the sparse feature maps, the estimated HSI is generated by

$$\widehat{\mathcal{X}} = \mathcal{W}_1^r \otimes \mathcal{W}_2^r \otimes \mathcal{W}_3^r \otimes \mathcal{S}_{K+1} \quad (18)$$

where $\{\mathcal{W}_1^r, \mathcal{W}_2^r, \mathcal{W}_3^r\}$ are also a set of learnable deconvolutional operations to reconstruct the clean HSIs.

Fig. 18 shows the network structure corresponding to (17) and (18). As can be seen, the unfolded network derived from (11) shares a very similar network architecture with DPNet-S, except that DPNet-S contains the ReLU activation function after each convolutional and deconvolutional operation, which is helpful to increase the representation ability of a network.

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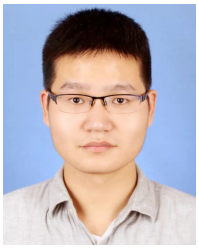
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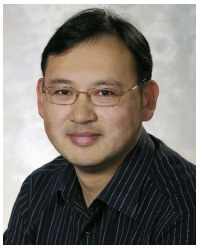
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Fengchao Xiong (Member, IEEE) received the B.E. degree in software engineering from Shandong University, Jinan, China, in 2014, and the Ph.D. degree from the College of Computer Science, Zhejiang University, Hangzhou, China, in 2019.

He visited Wuhan University, Wuhan, China, from 2011 to 2012, and Griffith University, Nathan, QLD, Australia, from 2017 to 2018. He is currently an Associate Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China. Besides this appointment, he is also a Post-Doctoral Fellow with the State Key Laboratory of Internet of Things for Smart City, Department of Computer and Information Science, University of Macau, Macau, China. His research interests include hyperspectral image processing, machine learning, and pattern recognition.

Dr. Xiong serves as a Topical Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Jun Zhou (Senior Member, IEEE) received the B.S. degree in computer science and the B.E. degree in international business from the Nanjing University of Science and Technology, Nanjing, China, in 1996 and 1998, respectively, the M.S. degree in computer science from Concordia University, Montreal, QC, Canada, in 2002, and the Ph.D. degree from the University of Alberta, Edmonton, AB, Canada, in 2006.

In June 2012, he joined the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia, where he is currently a Professor. Prior to this appointment, he was a Research Fellow with the Research School of Computer Science, Australian National University, Canberra, ACT, Australia, and a Researcher with the Canberra Research Laboratory, NICTA, Canberra. His research interests include pattern recognition, computer vision, and spectral imaging with their applications in remote sensing and environmental informatics.

Dr. Zhou is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Pattern Recognition* journal.



Jiantao Zhou (Senior Member, IEEE) received the B.Eng. degree from the Department of Electronic Engineering, Dalian University of Technology, Dalian, China, in 2002, the M.Phil. degree from the Department of Radio Engineering, Southeast University, Nanjing, China, in 2005, and the Ph.D. degree from the Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology, Hong Kong, in 2009.

He held various research positions at the University of Illinois at Urbana–Champaign, Champaign, IL, USA; The Hong Kong University of Science and Technology; and McMaster University, Hamilton, ON, Canada. He is currently a Professor with the Department of Computer and Information Science, Faculty of Science and Technology, University of Macau, Macau. He holds four granted U.S. patents and two granted Chinese patents. His research interests include multimedia security and forensics, multimedia signal processing, artificial intelligence, and big data.

Dr. Zhou has coauthored two papers that received the Best Paper Award at the IEEE Pacific-Rim Conference on Multimedia in 2007 and the Best Student Paper Award at the IEEE International Conference on Multimedia and Expo in 2016. He is serving as an Associate Editor for IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON MULTIMEDIA, and IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING.



Jianfeng Lu (Member, IEEE) received the M.S. and Ph.D. degrees in pattern recognition and intelligent system from the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China, in 1994 and 2000, respectively.

He is currently a Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology. His research interests include image analysis, machine learning, data mining, and intelligent robots.



Yuntao Qian (Senior Member, IEEE) received the B.E. and M.E. degrees in automatic control from Xi'an Jiaotong University, Xi'an, China, in 1989 and 1992, respectively, and the Ph.D. degree in signal processing from Xidian University, Xi'an, in 1996.

From 1996 to 1998, he was a Post-Doctoral Fellow with Northwestern Polytechnical University, Xi'an. Since 1998, he has been with the College of Computer Science, Zhejiang University, Hangzhou, China, where he became a Professor in 2002. He was a Visiting Professor with Concordia University, Montreal, QC, Canada, from 1999 to 2001; Hong Kong Baptist University, Hong Kong, in 2006; Carnegie Mellon University, Pittsburgh, PA, USA, in 2010; the Canberra Research Laboratory, NICTA, Canberra, ACT, Australia, in 2013; the University of Macau, Macau, from 2015 to 2016; and Griffith University, Nathan, QLD, Australia, in 2018. His research interests include machine learning, signal and image processing, pattern recognition, and hyperspectral imaging.

Dr. Qian is currently an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

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